DIN mounted Industrial switch (TSN-6325-8T4S4X)

A DIN-rail mounted industrial Layer-3 managed Ethernet switch is designed for reliable networking in harsh industrial environments. It typically features multiple Gigabit Ethernet ports along with high-speed fiber uplink ports to support large data traffic and fast communication. Built with a rugged metal enclosure, it can operate in wide temperature ranges and withstand vibration and electrical interference. Such switches support advanced networking functions like VLAN, QoS, redundancy protocols, precision time synchronization, and strong security features, making them suitable for industrial automation, smart manufacturing, transportation systems, and critical infrastructure networks.

Introduction

A DIN-rail mounted industrial Ethernet switch is a specialized networking device designed to provide reliable communication in harsh industrial environments. It is built with a rugged metal enclosure that can withstand extreme temperatures, vibration, dust, and electrical noise. These switches are typically installed inside control cabinets using a DIN rail mounting system, which allows quick, secure, and space-efficient installation in industrial panels.

Such switches play an important role in industrial networking by connecting devices like PLCs, sensors, controllers, and monitoring systems. They support stable and real-time data transmission along with advanced features such as network redundancy, traffic prioritization, and enhanced security. Because of their durability and reliability, they are widely used in industrial automation, manufacturing plants, power systems, and transportation infrastructure.

Features

🔌 **Physical Ports:**
- 8 × 10/100/1000BASE-T RJ45 copper ports
- 4 × 1G/2.5G SFP slots (auto-detect SFP type)
- 4 × 10G SFP+ slots (supports 10G / 2.5G / 1G)
- RJ45-to-RS232 console port for management

🏭 **Industrial Hardened Design:**
- Dual redundant power input (DC 9–48V or AC 24V)
- Reverse polarity protection & power failure backup
- DIN-rail & wall mounting support
- IP30 rugged aluminum enclosure
- ESD protection: 6000V DC
- Operating temperature: -40°C to 75°C

⚡ **Time-Sensitive Networking (TSN)**
- IEEE 1588 Precision Time Protocol (PTP)
- 802.1AS time synchronization (gPTP)
- 802.1Qbv scheduled traffic shaping
- 802.1Qch cyclic queuing
- 802.1Qci stream filtering & policing
- 802.1CB seamless redundancy (FRER)
- Frame preemption & express traffic support

🌐 **Layer 3 Routing Features:**
- IPv4 routing: RIPv2, OSPFv2
- IPv6 routing: OSPFv3
- Static IPv4/IPv6 routing
- VLAN-based routing interfaces

Why Choose Us?

🚀 Rapid integration, faster time-to-market

📘 Clean, verified, documented RTL

🤝 Direct access to experts

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Features Cont..

🔄 **Layer 2 Switching Features:**
- VLAN support (802.1Q, Q-in-Q, Private VLAN)
- STP / RSTP / MSTP support
- Link aggregation (LACP)
- Port mirroring & loop protection
- LLDP & UDLD support

🎯 **Quality of Service (QoS):**
- 8 priority queues per port
- Traffic classification (CoS, DSCP, TCP/UDP ports)
- Bandwidth control per port
- Strict priority & WRR scheduling

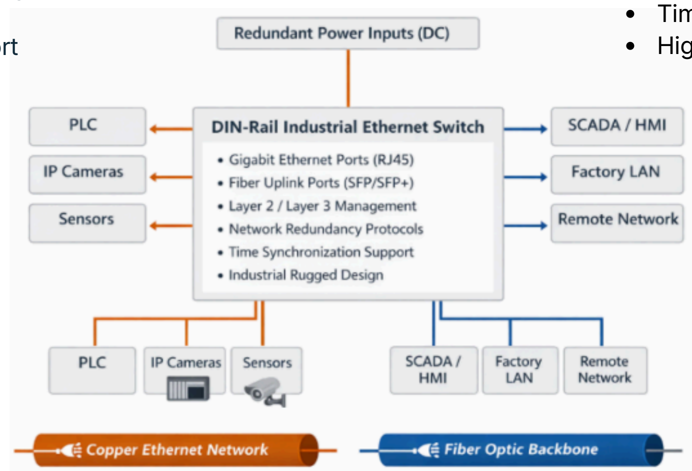
📡 **Multicast Features:**
- IGMP snooping (v1/v2/v3)
- MLD snooping for IPv6
- Multicast VLAN registration

🔒 **Security Features:**
- 802.1X authentication (port/MAC-based)
- ACL (IP & MAC based)
- DHCP snooping & IP source guard
- Dynamic ARP inspection
- RADIUS & TACACS+ support

🛠 **Management Features:**
- Web, CLI, SNMP management
- SSHv2, TLS, SNMPv3 secure access
- Dual firmware images
- DHCP server/relay support
- NTP, Syslog, SNMP traps

🚨 **Alarm & Monitoring:**
- Digital input/output (DI/DO)
- Power failure alarm relay
- Email & SNMP trap alerts
- Real-time SFP diagnostics

⚡ **Performance:**
- Switching fabric: 116 Gbps
- Throughput: 86.3 Mpps
- Jumbo frame: up to 10K bytes
- 📌 **Typical Applications:**
- Industrial automation networks
- 5G base stations
- Factory monitoring systems
- Time-critical control networks
- High-bandwidth fiber backbone



Specification

HARDWARE SPECIFICATIONS	
Parameter	**Specification**
Copper Ports	• 8 10/100/1000BASE-T RJ45 auto-MDI/MDI-X ports
SFP Port	• 8 100/1000/2500BASE-X SFP portsslot interfaces (Ports -9 to and Port-12) • Compatible with 100BASE-FX and 2500BASE-X SFP transceiver
SFP+ Ports	• 4 10GBASE-SR/LR SFP+ slots • Backward compatible with 100BASE-FX, 1000BASE-SX/LX/BX and 2500BASE-X SFP transceivers
Console	• 1 x RJ45-to-RS232 serial port (115200, 8, N, 1)

Reset Button	• < 5 sec: System reboot> 5 sec: Factory default
Connector	• Removable 6-pin terminal block for power input • Pin 1/2 for Power 1, Pin 3/4 for fault alarm, Pin 5/6 for Power 2 • Removable 6-pin terminal block for DI/DO interface • Pin 1/2 for DI 1 & 2, Pin 3/4 for DO 1 & 2, Pin 5/6 for GND
Alarm	• One relay output for power failure. Alarm relay current carry ability: 1A @ 24V DC
Digital Input (DI)	**2 digital input:** • Level 0: -24~2.1V (±0.1V) • Level 1: 2.1~24V (±0.1V) • Input load to 24V DC, 10mA max.
Digital Output (DO)	2 digital output: • Open collector to 24VDC, 100mA
Enclosure	• IP30 aluminum case
Installation	• DIN-rail or wall mounting
SDRAM	• 2048Mbytes
Flash Memory	• 64Mbytes
Dimensions (W x D x H)	• 86 x 135 x 152 mm
Weight	• 1,597g
Power Requirements	• DC 9~48V, 5A max. • AC 24V, 2A max.
Power Consumption	**DC input:** • Max. 16.8 watts/57.3BTU (system on) • Max. 38.2 watts/130.3BTU (Full loading) **AC 24V input:** • Max. 21.7 watts/74BTU (system on) • Max. 30 watts/102.3BTU (Full loading)
ESD Protection / Surge Protection	• 5KV DC / 6KV DC
LED Indicators	• **System:** Power 1 (Green), Power 2 (Green), Fault Alarm (Red), Ring (Green), Ring Owner (Green), DIDO (Red) • **Per 10/100/1000T RJ45 Port:** 1000Mbps LNK/ACT (Green), 10/100Mbps LNK/ACT (Amber) • **Per SFP Port:** 1G/2.5Gbps LNK/ACT (Green), 100Mbps LNK/ACT (Amber) • **Per SFP+ Port:** 1G/2.5Gbps LNK/ACT (Green), 10Gbps LNK/ACT (Amber)

SWITCHING SPECIFICATIONS	
Switch Architecture	• Store-and-Forward
Switch Fabric	• 116Gbps/non-blocking
Throughput	• 86.3Mpps@64Bytes
Address Table	• 32K entries, automatic source address learning and aging
Shared Data Buffer	• 32Mbits
Jumbo Frame	• 10K bytes
Flow Control	• IEEE 802.3x pause frame for full duplex • Back pressure for half duplex
LAYER 3 FUNCTIONS	
IP Interfaces	• Max. 128 VLAN interfaces
Routing Table	• Max. 512 static route entriesMax. 3072 routing table entries
Routing Protocols	• IPv4 RIPv2, IPv4 OSPFv2, IPv4 hardware static routing • IPv6 OSPFv3, IPv6 hardware static routing
LAYER 2 FUNCTIONS	
Port Configuration	• Port disable/enable • Auto-negotiation 10/100/1000Mbps full and half duplex mode selection • Flow control disable/enable, Port link capability control
Port Mirroring	• TX/RX/Both, Many-to-1 monitor • Mirror – Remote Switched Port Analyzer (Cisco RSPAN) • Supports up to 5 sessions
Port Status	• Display each port's speed duplex mode, link status, flow control status, auto-negotiation status, trunk status
VLAN	• IEEE 802.1Q tagged VLAN, IEEE 802.1ad Q-in-Q tunneling • Private VLAN Edge (PVE), MAC-based VLAN • Protocol-based VLAN, Voice VLANIP Subnet-based VLAN • MVR (Multicast VLAN registration), GVRP • Up to4K VLAN groups, out of 4095 VLAN IDs
Link Aggregation	• IEEE 802.3ad LACP/static trunk • 8 trunk groups with 16 ports per trunk group

Spanning Tree Protocol	• IEEE 802.1D Spanning Tree Protocol • IEEE 802.1w Rapid Spanning Tree Protocol • IEEE 802.1s Multiple Spanning Tree Protocol • Supports 7 MSTP instances • BPDU Guard, BPDU filtering and BPDU transparent, Root Guard
IGMP Snooping	• IPv4 IGMP (v1/v2/v3) snooping, IPv4 IGMP querier mode support • Supports 255 IGMP groups
MLD Snooping	• IPv6 MLD (v1/v2) snooping, IPv6 MLD querier mode support • Supports 255 MLD groups
Bandwidth Control	Per port bandwidth control • Ingress: 10Kbps~13128Mbps • Egress: 10Kbps~13128Mbps
Ring	• Supports ERPS, and complies with ITU-T G.8032 • Recovery time < 10ms @ 3 nodes • Recovery time <50ms @ 16 nodes • Supports Major ring and sub-ring
Synchronization	• IEEE 1588v2 PTP(Precision Time Protocol) • PTP Master, PTP Slave • Boundary clock • Peer-to-peer transparent clock, End-to-end transparent clock
QoS	• Traffic classification based, strict priority and WRR • 8-level priority for switching: Port number • 802.1p priority, 802.1Q VLAN tag • DSCP/ToS field in IP packet
Time-Sensitive Networking Protocols	**High Precision Time Synchronization** • IEEE1588 (Time Stamping) • 802.1AS-Rev gPTP default profile **Shapers** • 802.1Qbv (Time-aware Scheduling) • 802.1Qch (Cyclic Queuing and Forwarding) **TSN Stream Policing** • 802.1Qci (Per Stream Filtering and Policing) **Redundancy** • 802.1CB (Frame Replication and Elimination for Redundancy for seamless redundancy) • Also standard Linear and Ring protection **Delay Reduction** • EEE 802.1Qbu Frame Preemption,IEEE 802.3br Interspersing Express Traffic (IET)
SECURITY FUNCTIONS	
AAA	• RADIUS clientTACACS+ client

Access Control List	• IP-based ACL/MAC-based ACL • ACL based on: **1.** MAC Address **2.** IP Address **3.** Ethertype **4.** Protocol Type **5.** VLAN ID **6.** DSCP **7.** 802.1p Priority **8.** Up to 512 entries
Security	• Port security • IP source guard, up to 512 entries • Dynamic ARP inspection, up to 1K entries • Command line authority control based on user level • Static MAC address, up to 64 entries
Network Access Control	• IEEE 802.1x port-based network access control • MAC-based authentication • Local/RADIUS authentication
MANAGEMENT	
Basic Management Interfaces	• Console; Telnet; Web browser; SNMP v1, v2c
Secure Management Interfaces	• SSHv2, TLSv1.2, SNMPv3
System Management	• Firmware upgrade by HTTP protocol through Ethernet network • Configuration upload/download through HTTP • Remote Syslog • System log, LLDP protocol, NTP • PLANET Smart Discovery Utility, PLANET Cloud, Viewer app
STANDARDS CONFORMANCE	
SNMP MIBs	• RFC 1213 MIB-II, RFC 1493 Bridge MIB • RFC 1643 Ethernet MIB, RFC 2863 Interface MIB • RFC 2665 Ether-Like MIB, RFC 2819 RMON MIB (Group 1, 2, 3 and 9) • RFC 2737 Entity MIB, RFC 2618 RADIUS Client MIB • RFC 2863 IF-MIB, RFC 2933 IGMP-STD-MIB • RFC 3411 SNMP-Frameworks-MIB, RFC 4292 IP Forward MIB • RFC 4293 IP MIB, RFC 4836 MAU-MIB • IEEE 802.1X PAE, LLDP
Regulatory Compliance	• FCC Part 15 Class A • CE: **1.** EN55032 **2.** EN55035
Stability Testing	• EC60068-2-32 (free fall) • IEC60068-2-27 (shock) • IEC60068-2-6 (vibration)

Standards Compliance	• IEEE 802.3 10BASE-T, IEEE 802.3u 100BASE-TX/100BASE-FX • IEEE 802.3z Gigabit SX/LX, IEEE 802.3ab Gigabit 1000T • IEEE 802.3ae 10Gb/s Ethernet, IEEE 802.3bz 2.5GBASE-X • IEEE 802.3x flow control and back pressure, IEEE 802.3ad port trunk with LACP • IEEE 802.1D Spanning Tree Protocol, IEEE 802.1w Rapid Spanning Tree Protocol • IEEE 802.1s Multiple Spanning Tree Protocol, IEEE 802.1p Class of Service • IEEE 802.1Q VLAN tagging, IEEE 802.1X Port Authentication Network Control • IEEE 802.1ab LLDP, IEEE 802.3ah OAM • IEEE 802.1ag Connectivity Fault Management (CFM) • IEEE 802.1AS – Timing and Synchronization for Time-sensitive Applications • IEEE 802.1Qbu Frame Preemption,IEEE 802.3br Interspersing Express Traffic (IET) • IEEE 802.1Qci Per-Stream Filtering and Policing (PSFP) • IEEE 802.1Qbv Enhancements for Scheduled Traffic • IEEE 802.1CB Frame Replication and Elimination for Reliability (FRER) • RFC 768 UDP, RFC 783 TFTP, RFC 791 IP, RFC 792 ICMP, RFC 2068 HTTP • RFC 1112 IGMP v1, RFC 2236 IGMP v2, RFC 3376 IGMP v3, RFC 2710 MLD v1 • RFC 3810 MLD v2, RFC 2328 OSPF v2, RFC 5340 OSPF v3, RFC 2453 RIP v2 • ITU-T G.8032 ERPS Ring
ENVIRONMENT	
Operating	• -40 ~ 75 degrees C
Storage	• -40 ~ 85 degrees C
Humidity	• 5 ~ 95% (non-condensing)

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